MIL-P-1394F
23 November, 1983
SUPERSEDING
MIL-P-1394E
4 June 1974

# MILITARY SPECIFICATION

## PRIMER, ELECTRIC, M52A3B1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the metal parts and the loading, assembly, and packing of electric primer M52A3B1, primarily designed for use in the assembly of 20MM ammunition.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | | |
|---|---|---|
| MIL-A-48078 | - | Ammunition, Standard Quality Assurance Provisions, General Specifications for |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - | Inspection Terms and Definitions |
| MIL-STD-1168 | - | Lot Numbering of Ammunition |

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.7)

U.S. Army Armament Research & Development Center (ARDC)

Product and Packaging Drawings

| | | |
|---|---|---|
| 7258817 | – | Cartridge, 20MM, Target Practice, M55A2 |
| 7548057 | – | Cup, Primer |
| 7548058 | – | Button, Primer |
| 7548066 | – | Primer, Electric, M52A3B1 Assembly |
| LI 7548066 | – | Index of Inspection Equipment Lists for: Primer, Electric, M52A3B1 |
| 7548108 | – | Cup, Support, Primer |
| 8886458 | – | Box, Pallet Type, Wirebound, Packaging Assembly |
| 9329662 | – | Box, Set-up, Inner, Ammunition, Packing Assembly for Primer, Electric, M52A3B1 |
| 9329663 | – | Packing and Marking for Box, Metal, Ammunition M2A1 for Primer, Electric, M52A3B1 |
| 9329664 | – | Packing and Marking for Box, Wirebound, Packing Ammunition for Primer, Electric, M52A3B1 |

US ARMY MATERIAL DEVELOPMENT AND READINESS COMMAND

| | | |
|---|---|---|
| 19-48-4116-20PA1002 | – | Basic Procedures – Unitization Procedure for Boxed Ammunition and Components |
| 19-48-4116/159A-20PA1002 | - | Appendix 159A - Unitization Procedure for Boxed Ammunition and Components |



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Inspection Equipment Drawings

| | | |
|---|---|---|
| 7259300 | - | Tester, Bridge Thickness, for Electric Primers |
| 7259390 | - | Gage, Indicator and Snap |
| 7259427 | - | Gage, Indicator and Snap |
| 7259772 | - | Holding Fixture Primer Electric Sensitivity |
| 7259783 | - | Holding Fixture Primer Electric Sensitivity |
| 7259786 | - | Holding Fixture Primer Electric Sensitivity |
| 7288641 | - | Indicator Gage (Alternate) |
| 7296712 | - | Indicator Gage |
| 7553436 | - | Machine, Sensitivity, Testing, Small Arms Ammunition, Electric Primer Assembly |
| 7557339 | - | Go, Plain Ring Gage |
| 11075192 | - | Breechblock Assembly |
| 11075193 | - | Counter Chronograph |
| 11075194 | - | Oscilloscope |
| 11075227 | - | Pulse Firing Unit |
| 11075284 | - | Primer Resistance Test Circuit |

PUBLICATIONS

US ARMY MATERIEL COMMAND

| | | |
|---|---|---|
| AMCR 715-505 | - | Ammunition Ballistics Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges |

ARMAMENT COMMAND

| | | |
|---|---|---|
| ORD-SIP-S314 | - | Visual Inspection Standards for Small Arms Ammunition Primer Defects |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.



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INTERSTATE COMMERCE COMMISSION

49 CPR 71-90 - Rules and Regulations of the Transportation of Explosives and Other Dangerous Articles

The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1966 Edition and revision) available from the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402. Orders for the above publication, cite 49 CPR 71-90 (latest revision).

2.1.4 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General.

a Material. Materials shall be in accordance with the applicable drawings and specifications.

b. Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without prior approval (6.3).

3.2 Components. The components and assemblies shall comply with the requirements specified on Dwg. 7548066 (as applicable), referenced specifications and other requirements specified herein.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.4 Insulation strength. The cup-insulator-button assembly shall withstand an electrical potential as specified on Dwg. 7548066.

3.5 Electrical resistance. The electrical resistance of the primer assembly (dry) shall be as specified on Dwg. 7548066.

3.6 Pellet weight. The total dry weight of the primer mix shall be as specified on Dwg. 7548066.



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3.7 Electric primer sensitivity. The primer shall function when energized by a 10 microsecond discharge from a two microfarad condenser charged at one hundred sixty volts DC.

3.8 Electric primer time. The primer time shall not exceed .300 millisecond when energized by a 10 microsecond discharge pulse from a 2 microfarad condenser charged at 160 volts DC.

3.9 Action time. The action time of the test cartridge shall be as specified on Dwg. 7548066.

3.10 Function and casualty. The primer shall function without casualty in the test cartridge.

3.11 Contamination of energetic components.Extreme care shall be exercised to avoid contamination of primers by oil, grease or other foreign matter.

3.12 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.12.1 Processing defects. The metal parts and assemblies shall be free of cracks, splits ruptures, perforations, dents, deformities, exposed explosive and other forms of substandard workmanship.

3.12.2 Cleanliness. All components and assemblies shall be free of foreign matter, such as oil, grease, corrosion, dirt, chips, etc.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection (4.3)
b. Quality Conformance Inspection (4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.



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Dimensional and Test Inspections

| Part Description | Drawing | Quantity |
|---|---|---|
| Cup, primer | 7548057 | 125 |
| Button, primer | 7548058 | 125 |
| Cup, support, primer | 7548108 | 125 |
| Primer, Electric, M52A3B1 | 7548066 | 2000 |

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078. Failure of the original first article sample to be in accordance with the applicable drawings and requirements of Table I, as specified herein will result in sample disapproval.



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TABLE I. FIRST ARTICLE INSPECTION

**CLASSIFICATION OF DEFECTS & TESTS** MIL-P-1394F

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| | Primer, Electric, M52A3B1, and Components | 1 OF 1 | SEE BELOW |
| | | | NEXT HIGHER ASSEMBLY |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| | Cup, Primer - Dwg. 7548057<br>Examination of defects | 125 | | 3.2 | 4.4.2.1 |
| | Button, Primer - Dwg. 7548058<br>Examination of defects | 125 | | 3.2 | 4.4.2.2 |
| | Cup, Support, Primer - Dwg. 7548108<br>Examination of defects | 125 | | 3.2 | 4.4.2.3 |
| | Primer, Electric, M52A3B1 - Dwg. 7548066<br>Examination of defects and tests | 2000 | | 3.2 | 4.4.2.4/4.4.3 |
| | Packaging - Dwgs. 9329662, 9329663, 9329664<br>(As required by procuring agency)<br>Examination of defects | 2 | | 3.1 | 4.4.5 |

NOTES:

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4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. For primer and primer components, examination for major and minor defects as listed herein may be considered collectively on a class basis or defects may be considered individually. When a class basis is utilized the Acceptable Quality Level (AQL) for major defects shall be 1.50 percent and the AQL for minor defects shall be 2.50 percent. However, when major defects are considered collectively on a class basis, the AQL for any individual defect shall be limited to .40 percent. In addition, when three or less defects appear in a classification sampling plans applicable to a class shall not be used for major defects. (See 6.5)

c. For packing and marking, examination for major and minor defects shall be performed on an individual basis in accordance with the classification of defects. The AQL for the major defects shall be .40 percent and the AQL for the minor defects shall be .65 percent.



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d. Reference shall be made to ORD-SIP-S314 for judgement of degree of visual defectiveness of the primer and its components. Defect classifications of ORD-SIP-S314 are revised as shown by Table II.

TABLE II.

| Component | Defect | SIP Classification | Revised Classification |
|---|---|---|---|
| Cup | Lamination | Minor | Permissible |
| Cup | Dent | Minor | Permissible |
| Cup | V or U | Minor | Permissible |
| Cup | Stain or corrosion | Minor | Permissible |
| Cup | Scratch or nick | Major | Minor |
| Button | Dent | Minor | Major |
| Button | Stain or corrosion | Minor | Permissible |
| Button | Scratch or nick | Minor | Major |
| Support Cup | Stain or corrosion | Minor | Permissible |
| Support Cup | V or U | Minor | Permissible |
| Support Cup | Flash hole missing or obstructed | Minor | Major |
| Support Cup | Edge crooked or eared | Minor | Major |
| Primer Assy | Inverted button | Major | Critical |
| Primer Assy | Missing, misplaced or protruding insulator | Minor | Major |
| Primer Assy | Foreign matter other than oil or grease | Minor | Major |



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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-P-1394F

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.1 | Cup, Primer | | | | 7548057 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | 7548066 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| CRITICAL | None defined | | | | |
| MAJOR | | | | | |
| 101. | Cracks, split, or lamination | | .25% | 3.2 | Visual |
| 102. | Dent, ragged edge (V or U) | | .25% | 3.2 | Visual |
| 103. | Wire-like edge, or slivers lodged in or attached to cup | | .25% | 3.2 | Visual |
| 104. | Foreign matter, stain or corrosion | | .25% | 3.12 | Visual |
| MINOR | | | | | |
| 201. | Height | | .40% | 3.2 | Gage |
| 202. | Diameter, outside | | .40% | 3.2 | Gage |
| 203. | Scratch or nick | | .40% | 3.2 | Visual |
| 204. | Evidence of poor workmanship | | .65% | 3.12 | Visual |

NOTES:

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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.2 | Button, Primer | 1 OF 1 | 7548058 |
| | | | NEXT HIGHER ASSEMBLY: 7548066 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| CRITICAL | None defined | | | | |
| MAJOR | | | | | |
| 101. | Improperly formed | | .25% | 3.2 | Visual |
| 102. | Crack, split or lamination | | .25% | 3.2 | Visual |
| 103. | Dent | | .25% | 3.2/ 3.12 | Visual |
| 104. | Scratch or nick | | .25% | 3.2/ 3.12 | Visual |
| 105. | Foreign matter, stain or corrosion | | .25% | 3.2/ 3.12 | Visual |
| MINOR | | | | | |
| 201. | Overall height | | .40% | 3.2 | 7259390 |
| 202. | Outside diameter | | .40% | 3.2 | 7259390 |
| 203. | Web thickness | | .40% | 3.2 | 7259390 |
| 204. | Evidence of poor workmanship | | .65% | 3.12 | Visual |

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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 7548108 |
|---|---|---|---|---|---|
| 4.4.2.3 | Cup, Support, Primer | | | | NEXT HIGHER ASSEMBLY 7548066 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| CRITICAL | None defined. | | | | |
| MAJOR | | | | | |
| 101. | Ragged edges (V or U) | | .25% | 3.2 | Visual |
| 102. | Flash hole missing or obstructed | | .25% | 3.2 | Visual |
| 103. | Edge crooked or eared | | .25% | 3.2 | Visual |
| 104. | Burr at flash hole | | .25% | 3.2 | Visual |
| 105. | Foreign matter, stain or corrosion | | .25% | 3.2/ 3.12 | Visual |
| MINOR | | | | | |
| 201. | Height | | .40% | 3.2 | 7259427 |
| 202. | Outside diameter | | .40% | 3.2 | 7259427 |
| 203. | Crack, split or lamination | | .40% | 3.2/ 3.12 | Visual |
| 204. | Scratch, nick or dent | | .40% | 3.2 | Visual |
| 205. | Evidence of poor workmanship | | .65% | 3.12 | Visual |

NOTES:

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## CLASSIFICATION OF DEFECTS & TESTS

| PARAGRAPH | TITLE | | SHEET 1 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.4 | Primer, Electric, M52A3B1 1/ | | | | 7548066 |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| CRITICAL | | | | | |
| 1. | Missing, inverted or insecure support cup | | 100% | 3.2 | Visual/Manual |
| 2. | Oil or grease | | 100% | 3.2 | Visual |
| 3. | Inverted button | | 100% | 3.2 | Visual |
| 4. | Mixed types | | 100% | 3.2 | Visual |
| MAJOR | | | | | |
| 101. | Crushed or mutilated | | .25% | 3.2 | Visual |
| 102. | Missing, misplaced or protruding insulator | | .25% | 3.2 | Visual |
| 103. | No button or insulator covering button | | .25% | 3.2 | Visual |
| 104. | Presence of metal slivers on outside of primer or bridging the primer cup and button across insulator | | .25% | 3.2 | Visual |
| 105. | Foreign matter other than oil or grease | | .25% | 3.12 | Visual |

NOTES:
1/ Test requirements shall be in accordance with 4.4.3.

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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | SHEET 2 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.4 | Primer, Electric, M52A3B1 1/ | | | | 7548066 |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| MINOR | | | | | |
| 201. | Height | | .40% | 3.2 | Gage |
| 202. | Diameter | | .40% | 3.2 | Gage/7557339 |
| 203. | Depth, button | | .40% | 3.2 | 7288641/7296712 |
| 204. | Missing, slipped or punctured disc | | .40% | 3.2 | Visual |
| 205. | Cocked support cup | | .40% | 3.2 | Visual |
| 206. | Explosive composition on disc or support cup | | .40% | 3.2 | Visual 2/ |
| 207. | Evidence of poor workmanship | | .65% | 3.12 | Visual |

NOTES:

2/ Minute particles of explosive material are not considered cause for rejection.

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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

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| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.5 | Box, Set-up, Inner, Ammunition, Packing Assembly for, Primer, Electric, M52A3B1 (After Loading) | | 9329662 |
| | | | NEXT HIGHER ASSEMBLY 9329663 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| CRITICAL | None defined. | | | | |
| MAJOR | | | | | |
| 101. | Improper quantity of primers | | .40% | | Visual |
| 102. | Sleeve torn or ripped | | .40% | | Visual |
| 103. | Label missing or insecurely applied | | .40% | | Visual |
| 104. | Marking missing, incorrect or illegible | | .40% | | Visual |
| MINOR | | | | | |
| 201. | Improper positioning of primers | | .65% | | Visual |
| 202. | Evidence of poor workmanship | | .65% | | Visual |

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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.6 | Packing and Marking for Box, Metal Ammunition, M2A1 for Primer, Electric, M52A3B1 | | 9329663 |
| | | | NEXT HIGHER ASSEMBLY 9329664 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| CRITICAL | None defined. | | | | |
| MAJOR | | | | | |
| 101. | Incorrectly closed and secured box | | .40% | | Visual |
| MINOR | | | | | |
| 201. | Contents, incorrect (prior to closing) | | .65% | | Visual |
| 202. | Filler(s) missing or improper (prior to closing) | | .65% | | Visual |
| 203. | Severe dents or buckles | | .65% | | Visual |
| 204. | Marking missing, incorrect or illegible | | .65% | | Visual |
| 205. | Tag, marking missing, incorrect or illegible (interplant shipment only) | | .65% | | Visual |
| 206. | Evidence of poor workmanship | | .65% | | Visual |

NOTES:

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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

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| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.7 | Packing and Marking for Box, Wirebound Packing Ammunition for Primer, Electric, M52A3B1 | 1 OF 1 | 9329664 |
| | | | NEXT HIGHER ASSEMBLY |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| CRITICAL | | | | | |
| 1. | "Small Arms Primers-Handle Carefully" marking missing, incorrect or illegible | | 100% | | Visual |
| MAJOR | | | | | |
| 101. | Missing box end | | .40% | | Visual |
| 102. | Missing or incorrectly attached metallic seal | | .40% | | Visual |
| 103. | Unfastened binding wire | | .40% | | Visual |
| MINOR | | | | | |
| 201. | Filler(s) or separator(s) missing or improper (prior to closing) | | .65% | | Visual |
| 202. | Other marking missing, incorrect or illegible | | .65% | | Visual |
| 203. | Evidence of poor workmanship | | .65% | | Visual |

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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.8 | Unitization Procedures for Boxed Ammunition and Components (Palletized Unit Load) | | | | 19-48-4116/159A-20PA1002 |
| | | | | | NEXT HIGHER ASSEMBLY 19-48-4116-20PA1002 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | See drawings 19-48-4116-20PA1002 and 19-48-4116/159A-20PA/1002 | | | | |

NOTES:

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4.4.3 Tests. The tests listed in Table III shall be performed on each primer or each lot, as indicated, in accordance with the methods of inspection prescribed in 4.5. For acceptance, the test sample primers shall be selected at random from the lot quantity and shall be examined in accordance with 4.4.2. After examination, the sample primers shall be thoroughly mixed before being divided into samples for the various tests. Acceptance criteria for each test shall be as specified.

TABLE III

| Test | | SAMPLE SIZE 1st Art. | SAMPLE SIZE Prod. Lot | Test Equipment Number | | Rqmt/Test Para |
|---|---|---|---|---|---|---|
| Insulation strength | 1/ | All | All | Gage | | 3.4/4.5.1 |
| Electrical resistance | 2/ | All | All | 11075284 | | 3.5/4.5.2 |
| Minimum pellet weight | 3/ | All | All | 7259300 | | 3.6/4.5.3 |
| Electric primer sensitivity | 4/ | 800 | 800 | 7553436 | 8/, 9/ | 3.7/4.594 |
| Electric primer time | 5/ | 50 | 50 | 11075192<br>11075193, 11075194<br>11075227 | 8/ | 3.8/4.5.5 |
| Action time -65°F | 6/ | 50 | 50 | | | 3.9/4.5.6 |
| Function and casualty | 7/ | 1100 | 300 | | | 3.10/4.5.7 |

1/ A cup-insulator-button assembly which fails to comply with the insulation strength requirement shall be rejected.

2/ A primer which fails to comply with the applicable electrical resistance requirement shall be rejected.

3/ A primer which fails to comply with the applicable minimum dry weight requirement shall be rejected.

4/ If the average critical voltage (V) plus three standard deviations (3. ) exceeds 160 volts, the lot shall be rejected subject to retest. If on retest, V + 3 exceeds 160 volts the lot shall be rejected. If in either the first test or retest, a primer fails to fire at 160 volts, the lot shall be rejected.

5/ If the sample average primer time plus four standard deviations exceeds. 300 millisecond or if an individual primer time of the test sample exceeds. 300 millisecond, the lot shall be rejected.



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6/ If the sample average action time plus four standard deviatons exceeds 3.5 milliseconds, or an individual primer action time of the test sample exceeds 3.5 milliseconds, the lot shall be rejected.

7/ If a primer button is missing, a misfire occurs, or nine or more primers show evidence of leakage, as defined in AMCR 715-505, Volume 8, the lot shall be rejected. If more than three but less than nine primers show evidence of leakage, a second sample of double the number specified by Table III (Production Lot Only) shall be tested. If, in the accumulated samples, a button is missing, a misfire occurs, or nine or more primers show evidence of leakage, the lot shall be rejected.

8/ Principal assembly drawing no. only, detail drawings supplied upon request to procuring activity.

9/ Use applicable holding fixture as indicated below in lieu of 7553437 and 7553438 as shown on 7553436.

a. Holding fixture, primer 7259772

b. Holding fixture, primed case (M103) 7259703

c. Holding fixture, primed case (M21A1) 7259786

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Index of Inspection Equipment List-LI 7548066 and the “Paragraph Reference/Inspection Method” column in the tables starting with paragraph 4.4.2. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.2 herein.

4.4.5 Packaging,Packing, Marking and Palletization.

When shipment is required, packaging, packing, marking and palletization inspection shall comply with the requirements of paragraphs 4.4.2.5, 4.4.2.6, 4.4.2.7 and 4.4.2.8.

4.5 Methods of inspection.

4.5.1 Insulation strength. The insulation strength of the cup-insulator-button assembly shall be determined applying the specified current, voltage and timing.

4.5.2 Electrical resistance. Each primer shall be tested for the electrical resistance limits of 3.5. The inspection equipment limits the current and voltage to 1.4 milliamperes and 7.5 volts.



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4.5.3 Pellet weight. The minimum total dry weight of the primer mix shall be determined by measuring the recompressed bridge thickness of, the primer. Bridge thickness is defined as the dimension between the center of the outside face of the primer button and the disc, or between the center of the outside face of the primer button and the exposed face of the primer support cup. The dry primer mix in the primer assembly shall be recompressed prior to measuring the bridge thickness using force of approximately 100 pounds in excess of the pressure required to move the cup support with respect to the primer cup. The minimum bridge thickness dimension used to assure minimum total dry weight of the primer mix and the recompression force shall be established by the supplier and approved by the procuring activity.

4.5.4 Electric primer sensitivity. The method of test shall be as specified in AMCR 715-505, Volume 8. The primer firing device shall be adjusted for a 10 microsecond discharge time from a 2 microfarad condenser. The electrical sensitivity limit of the sample primers shall be determined using the specified test method and procedure.

4.5.5 Electric primer time. The method of test shall be as specified in AMCR 715-505, Volume 8. The test equipment shall be adjusted to comply with the specified current and voltage requirements.

4.5.6 Action time. The method of test shall be as specified in AMCR 715-505, Volume 8. The primer firing assembled as Cartridge, 20MM, Target Practice, M55A2, in accordance with 4.5.7. The test cartridges shall be conditioned at -65°F for a period of not less than four hours after the conditioning chamber has stabilized at -65°F ± 5°F. Timing for the conditioning period shall start after the chamber has stabilized following the sample being placed in the conditioning chamber.

4.5.7 Function and casualty. The method of test shall be as specified in AMCR 715-505, Volume 8. The test sample primers shall be inserted into cases that have been previously inspected for depth and diameter of primer pocket and length to shoulder basic diameter. The sample primed cases shall be assembled as Cartridges, 20MM, Target Practice, M55A2, conforming to Dwg. 7258817. Prior to firing, the test cartridges shall be inspected for depth of primer seating, primer crimp missing, loose primer, metal slivers on case head or primer, profile and alignment max., and presence of foreign matter on the primer button. The sample test cartridges at ambient temperature shall be fired in bursts of 50 cartridges at a cyclic rate not less than 1400 shots per minute in Gun, Automatic, M39. The gun barrel shall be at ambient temperature at the beginning of each burst.



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4.5.8 Misfires. Misfiring test cartridges shall be retested in the test equipment specified in EL 11075190, a detail of LI7548066. Test cartridge(s) which fail to fire using the specified equipment shall be classified as to the cause of the failure and recorded as misfire(s) in accordance with Table III.

5. PACKAGING

5.1 Preservation and Packaging.

5.1.1 Levels A and B. Packaging shall be in accordaance with Dwg. 9329662 and 9329663.

5.1.2 Level C (Interplant - CONUS).Same as levels A and B except containers shall be marked with a tag per Dwg. 9329663.

5.2 Packing.

5.2.1 Levels A and B. Packing shall be in accordance with Dwg. 9329664.

5.2.2 Level C. Not applicable.

5.3 Palletization.

5.3.1 Levels A and B. Palletize 36 wirebound boxes in accordance with Dwg. 19-48-4116/159A-20PA1002.

5.3.2 Level C (Interplant - CONUS). Palletize in accordance with Dwg. 8886458.

5.4 Marking.

5.4.1 Levels A and B. Mark in accordance with Dwgs. 9329662, 9329663 and 9329664.

5.4.2 Level C (Interplant - CONUS). Mark in accordance with Dwgs. 9329662, 9329663 (tag) and 8886458 except pallet marking of Dwg. 8886458 to be as follows:

NSN and DODIC
Nomenclature: “419328 (or lesser qty.) - Primer, Electric, M52A3B1”
Lot No.
Gross Weight
DOT: “Small Arms Primers - Handle Carefully”



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6. NOTES

6.1 Ordering data. See MIL-A-48078.

6.2 Submission of inspection equipment design for approval. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Center, ATTN: DRSMC-QAF-I(D), Dover, NJ, 07801.

6.3 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item.

6.4 First article sample. In order to minimize inspection costs, the firing tests specified in 4.4.3 will be performed after the sample has been found acceptable for all other requirements. Additional primers may be required by the test facility.

6.5 Intermediate point inspection. The classification of defects identifies the defect characteristics for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality of the end time.

6.6 Safety data. For safety data requirements refer to Safety Data Document as listed on Product Drawings.

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or U.S. Army Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities are now under the cognizance of ARDC.

6.8 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness-of the changes.

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Review activities:
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Preparing Activity:
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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-P-1394F (AR) | Primer, Electric, M52A3B1 |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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